

Reservoir Architecture

Online sequence learning

A temporal memory, consisting of a simple recurrent delay node and a hetero-associative predictor node. This model learns higher-order sequences and next-token prediction on the fly.

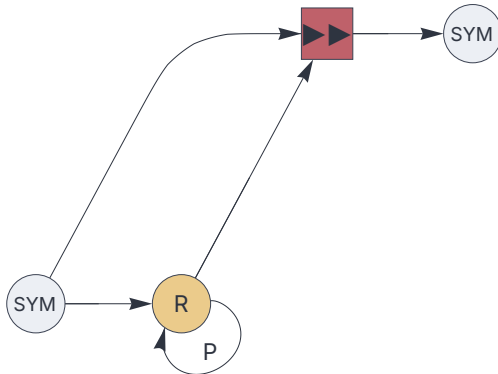
Requirements

```
<< "Circuits.m"
```

Circuit

```
{
  "$schema": "https://creatingintelligence.org/schemas/circuit-v1.json",
  "hyperparameters": {"default": [2000, 10]},
  "dataflow": [
    {"component": "input", "plugin": "codec", "send": [11]},
    {"component": "delay", "send": [51], "receive": [[11, {"slot": 51, "tags": ["permute"]}]], "rate_limit": 6.0},
    {"component": "predictor", "send": [101], "receive": [11, 51]},
    {"component": "output", "plugin": "codec", "receive": [101]}
  ]
}
```

```
circuit = Circuit[json];
c = circuit["function"];
circuit["schematics"][ImageSize->270]
```



Dataset

Generate a 2000 character random string.

```
chars = CharacterRange["A", "Z"]
```

```
{A, B, C, D, E, F, G, H, I, J, K, L, M, N, O, P, Q, R, S, T, U, V, W, X, Y, Z}
```

```
dataset = RandomChoice[ chars, 2000];
```

Prediction score

```
Do[
  predictions = c /@ dataset;
  Print["  iteration = ", i,
    ", prediction score = ", N[Mean@Boole@MapThread[SameQ,
      {dataset, RotateRight[predictions]}]]],
    {i, 1, 10}];
```

iteration = 1, prediction score = 0.027
iteration = 2, prediction score = 0.7965
iteration = 3, prediction score = 0.9575
iteration = 4, prediction score = 0.9925
iteration = 5, prediction score = 0.9985
iteration = 6, prediction score = 0.999
iteration = 7, prediction score = 0.9985
iteration = 8, prediction score = 0.999
iteration = 9, prediction score = 0.999
iteration = 10, prediction score = 1.